ELEC60029 Semiconductor Devices- Summer Resit 2023: *Answers:*

Q1

(a)

(i)

$n_i = p_i$ intrinsic hole concentration is equal to intrinsic electron concentration.

[2]

(ii)

$N_A =$ 2×10^{23} m^{-3} acceptor concentration

$p = p_i + N_A \simeq N_A$ (n_i is only $\sim 10^{16}\ m^{-3}$)

$p = N_A =$ $2 \times 10^{23}\ m^{-3}$

Or, $=$ $2 \times 10^{23}\ (10^2\ cm)^{-3}$

$=$ $2 \times 10^{23}\ 10^{-6}\ cm^{-3}$

$=$ $2 \times 10^{17}\ cm^{-3}$

$n = \frac{n_i^2}{N_A} = \left(\frac{1.45\ 10^{10}\ 10^6}{2 \times 10^{23}}\right)^2 \simeq 10^9\ m^{-3}$

Thus $n << p$

n: minority carrier density (electrons)

p: majority carrier density (holes)

[2]

(iii)

definition of conductivity

$$\sigma = e\,n\,\mu_n + e\,p\,\mu_p$$

unit charge; density of electrons; electron mobility; density of holes; hole mobility

in notes you'll find that $\mu_n >> \mu_p$ but $p >>> n$

thus $\sigma \simeq ep\mu_p$

$$\mu_p = \frac{\sigma}{ep} = \frac{1550}{2 \times 10^{23}\ m^{-3}\ 1.6\ 10^{-19} C} = 0.048\ \frac{m^2}{Vs}$$

(iv) [4]

$\mu_n \simeq 5\mu_p$ & $p \simeq 10^{14}\,n$

$$\Rightarrow \sigma = e\left(\frac{p}{10^{14}} \cdot 5\mu_p + p\mu_p\right) = e\left(\underbrace{\frac{5}{10^{14}} + 1}_{\simeq 1}\right)p\mu_p$$

$$= ep\mu_p$$

[4]

(b)

Increasing donor doping will increase the concentration of free electrons, *n*. This is because, assuming complete ionisation of the donor atoms, $n = N_D$. Mobility will decrease due to increased impurity scattering. If the number of scattering events increases, then the time between collisions will decrease, as the mobility is directly proportional to the time between collisions of the carriers with the impurity atoms.

[4]

(c)

(i). $@t < 0; i_p(0^-) = 0; Q_p(0^-) = 0$

$@t = 0^+; i_p(0^+) = I_{ON} = \frac{V}{R}$

$$I_{ON} = \frac{Q_p(t)}{\tau_p} + \frac{dQ_p(t)}{dt}$$

[3]

Integrating this equation to find Q_p gives:

$$Q_p(t) = I_{ON}\tau_p + Aexp\left(-\frac{t}{\tau_p}\right)$$

Introducing boundary conditions for $t=0^+$ allows us to find constant $A = -I_{ON}\,\tau_p$, giving:

ANSW

$$Q_p(t) = \tau_p I_{ON}\left(1 - \exp\left(-\frac{t}{\tau_p}\right)\right)$$ [3]

(ii) Similar to (i) above, but with revised boundary conditions, gives us:

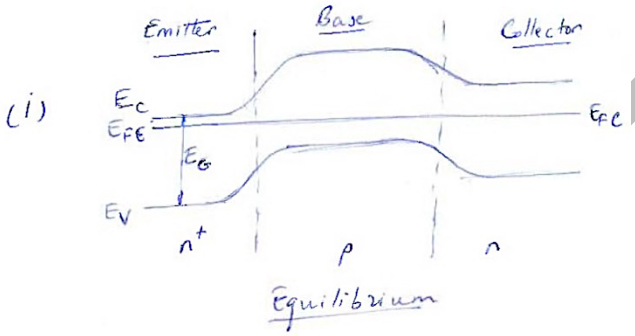
$$@t < 0; i_p(0^-) = I_{ON} = \frac{Q_p(0^-)}{\tau_p} \approx \frac{V}{R}$$

$$@t = 0; i_p(0) = 0$$

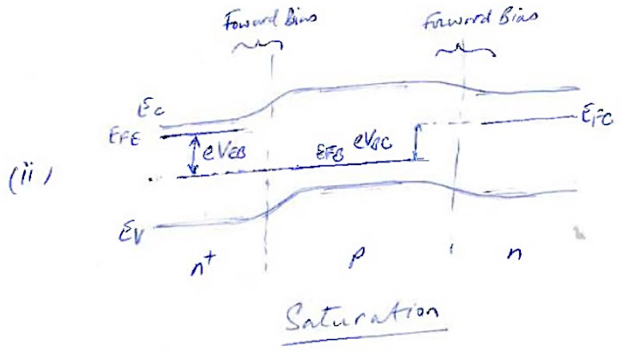
$$0 = \frac{Q_p(t)}{\tau_p} + \frac{dQ_p(t)}{dt}$$

$$Q_p(t) = I_{ON}\tau_p \exp\left(\frac{-t}{\tau_p}\right)$$ [6]

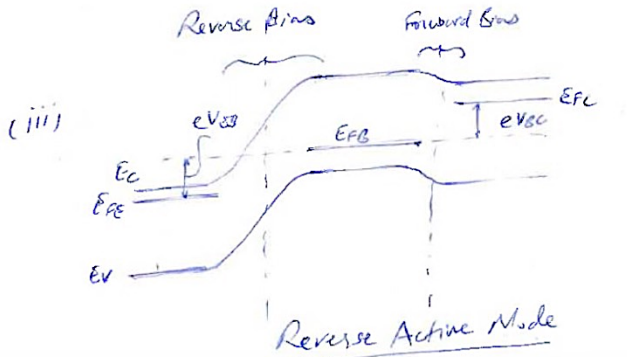
(d)



[4]



[4]



[4]

Q2

(a)

Each impact ionization event increases the carrier concentration by 2.

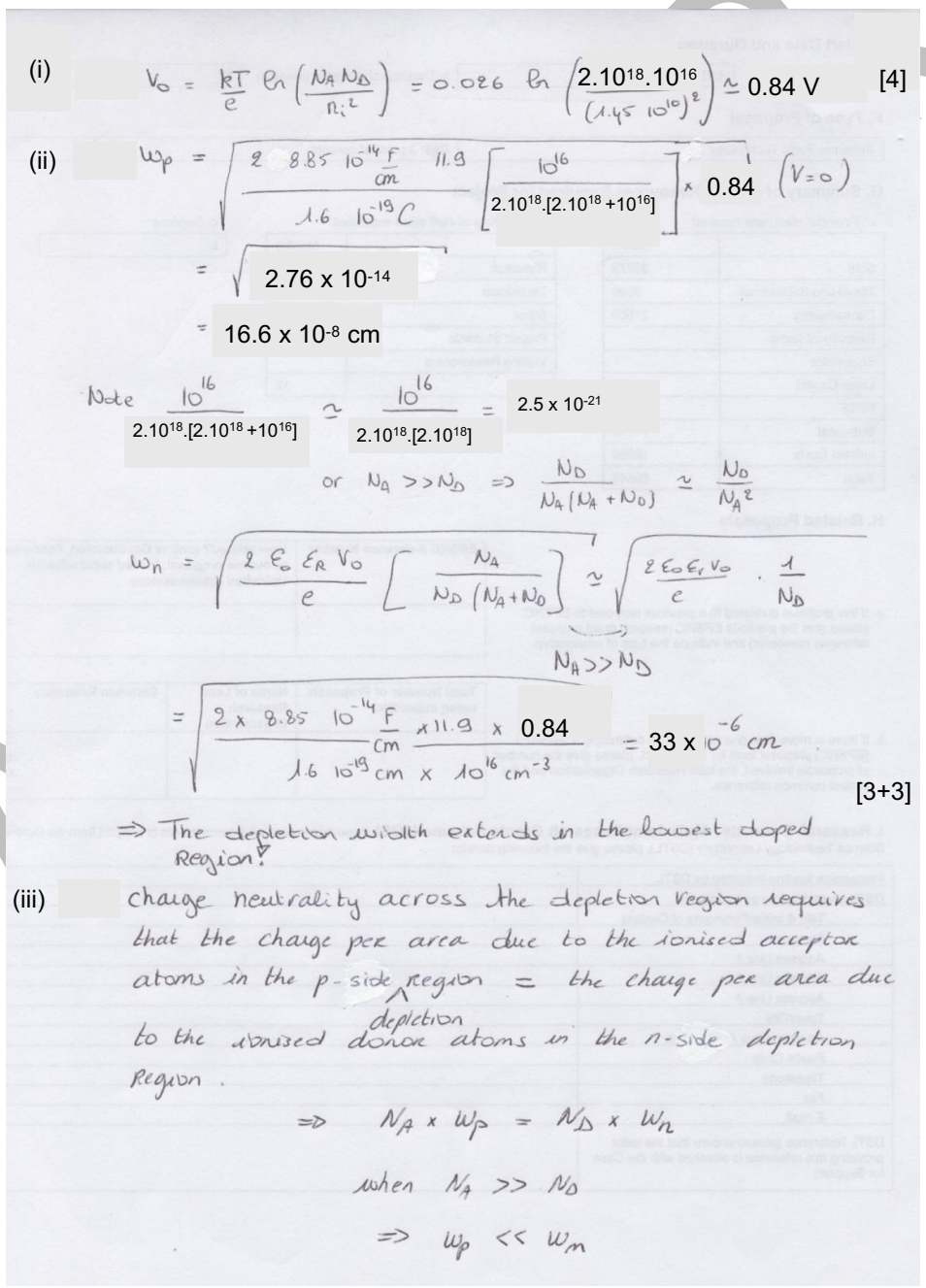
If the density of injected electrons into the depletion region in N_1, then the 1st ionization event increases this to 2 N_1, the second increases this to 4 N_1 etc. Hence, for m ionization events, the carrier density becomes $N_m = 2^m N_1$.

[3]

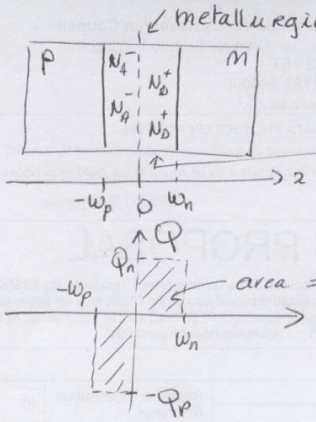
As the depletion width is 20 times the scattering length, $m = 20$. This gives the required ratio, $N_m / N_1 = 2^m = 2^{20} = 1 \times 10^6$.

[3]

(b)



(iii)



$$\left.\begin{aligned} Q_n &= e N_D w_n \cdot A \\ -Q_p &= -e N_A w_p A \end{aligned}\right\} \qquad A = \pi \left(\frac{d}{2}\right)^2 \qquad d = 10^{-2}\,cm$$

$$Q_n = 1.6\ 10^{-19}\,C \times 10^{16}\,cm^{-3}\ 33 \times 10^{-6}\,cm\ \pi \left(\frac{10^{-2}}{2}\right)^2 cm^2$$

$$= 4.15 \times 10^{-12}\,C$$

$$Q_p = +Q_n = +\,4.15 \times 10^{-12}\,C \quad \text{charge neutrality.}$$

[3+3]

(iv) Write 1D Poisson equation in the n-type depletion region

$$\frac{dE}{dx} = \frac{e N_D}{\varepsilon_0 \varepsilon_R} \quad \left(= \frac{\text{charges}}{\varepsilon} \text{ in } 0 < x < x_n\right)$$

(E: electric field)

$$\int_{E_0}^{0} dE = \int_{0}^{w_n} \frac{e N_D}{\varepsilon_0 \varepsilon_R}\, dx$$

[4]

E-field = minimum @ $x = 0$

$\cong 0$ outside depletion region

$$E_0 = -\frac{e N_D w_n}{\varepsilon_0 \varepsilon_R} = \frac{-1.6\ 10^{-19}\,C \times 10^{16}\,cm^{-3} \times 33\ 10^{-6}\,cm}{8.8\ 10^{-14}\,\frac{F}{cm} \times 11.9}$$

$$= -50126\ \frac{V}{cm}$$

[4]

Q3:

a)

i) $I_E = I_B + I_C$

[4]

ii)

$$I_E = eAn_i^2\left(\frac{D_{nE}}{(X_E - w_E)N_{A_E}} + \frac{D_{pB}}{(X_B - w_{Ef} - w_{Er})N_{D_B}}\right)\left(\exp\left(\frac{V_{EB}}{V_T}\right) - 1\right)$$

$$I_B = eAn_i^2\left(\frac{D_{nE}n_{P0_E}}{(X_E - w_E)N_{A_E}}\right)\left(\exp\left(\frac{V_{EB}}{V_T}\right) - 1\right)$$

$$I_C = eAn_i^2\left(\frac{D_{pB}}{(X_B - w_{Ef} - w_{Er})N_{D_B}}\right)\left(\exp\left(\frac{V_{EB}}{V_T}\right) - 1\right)$$

This ignores the reverse bias current from collector into base.

[10]

b) The transconductance g_m is the derivative of the output current to the controlled bias voltage. In BJTs output current is I_C and control bias voltage is V_{EB}.

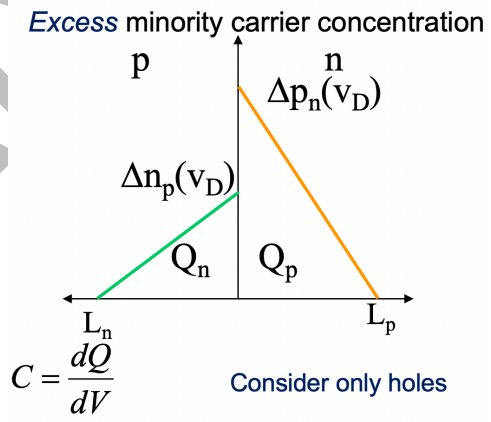
$$g_m = \frac{dI_C}{dV_{EB}} = \frac{eAn_i^2}{V_T}\left(\frac{D_{pB}}{(X_B - w_{Ef} - w_{Er})N_{D_B}}\right)\left(\exp\left(\frac{V_{EB}}{V_T}\right)\right)$$

$$g_m \approx \frac{I_C}{V_T}$$

[6]

c)

The excess, diffusion minority carrier concentration in the emitter (p-type) and base (n-type), near the EB junction, is shown below:



[5]

Hence, using diffusion capacitance $C = \mathrm{d}Q/\mathrm{d}V$ and Q_p, the hole excess charge (as $Q_p >> Q_n$, given the p^+ emitter), we obtain:

ANSWERS

$$C = \frac{dQ}{dV}$$

$$C_{diff} = \frac{dQ_p}{dV} = \frac{d\left(eAL_p \Delta p_n \big/ 2\right)}{dV} = \frac{eAL_p}{2} \frac{d\left(p_{n_0}\left(\exp\left(\frac{eV}{kT}\right) - 1\right)\right)}{dV} = \frac{eAL_p p_{n_0} \exp\left(\frac{V}{V_T}\right)}{2V_T}$$

$$C_{diff} \approx \frac{L_p^2 I_{Fp}}{2D_p V_T}$$

[5]

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Q4:

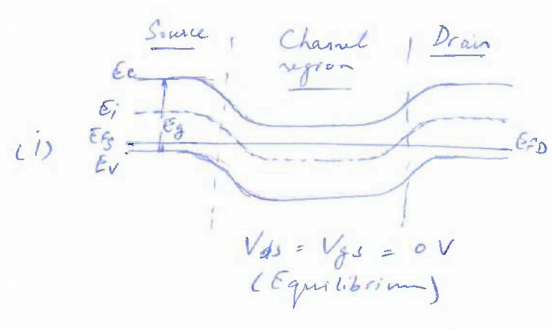
(a) (i) Source and drain: Doped p^{+} [2]

Substrate: Doped n, or n^{-} [2]

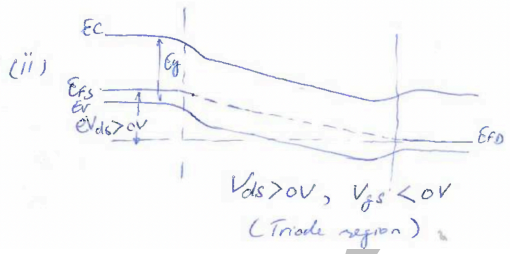
(ii) $V_{th} < 0$ (p-channel, enhancement mode). [2]

(iii) $0 > V_{gs} > V_{th.}$ [2]

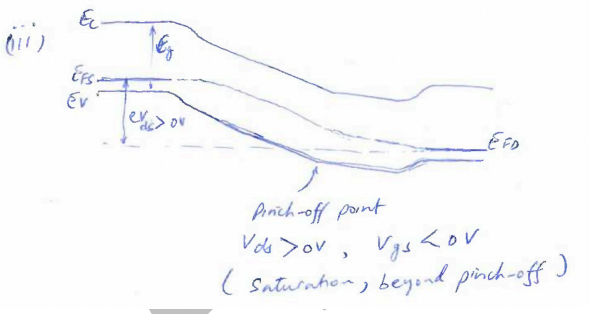
(b)



[2]



[2]



[4]

ANSWERS

(c)

$$I_{DS} = \frac{\mu_n C_{ox} W_g}{L_g}\left[(|V_{GS}|-|V_{TH}|)|V_{DS}| - \frac{V_{DS}^2}{2}\right]$$

Triode region:

$V_{DS} << V_{GS}-V_{TH} \Rightarrow I_{DS} \approx \frac{\mu_n C_{ox} W_g}{L_g}(|V_{GS}|-|V_{TH}|)|V_{DS}|$

$\Rightarrow g_m = \frac{\partial I_{DS}}{\partial V_{GS}} = \frac{\mu_n C_{ox} W_g}{L_g}|V_{DS}|$ [4]

Saturation region:

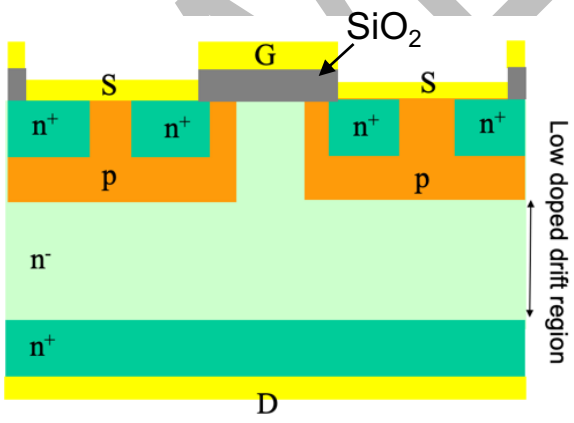
$V_{DS} = V_{GS}-V_{TH} \Rightarrow$ onset of saturation

$\Rightarrow I_{DS} = \frac{\mu_n C_{ox} W_g}{2L_g}(|V_{GS}|-|V_{TH}|)^2$

$\Rightarrow g_m = \frac{\partial I_{DS}}{\partial V_{GS}} = \frac{\mu_n C_{ox} W_g}{L_g}(|V_{GS}|-|V_{TH}|)$

[4]

(d)



[6]

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